

Economic Transformation

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Summary

Most FGD groups reported that CFM had not improved access to natural resources for either household use or sale, though approximately one third indicated that access had increased. Approximately half of FGD groups agreed that CFM had generated new employment opportunities, with a marked contrast between community youth (who were most likely to agree) and community female FGDs (who were most likely to disagree). Employment was strongly male-biased and the dominant employment category was HFO or Community Scout. Income earning opportunities from CFM remain limited for most communities. Awareness of carbon credit agreements was generally high. However, although around 80% of CFMGs are involved in generating carbon credits, only around one third had received carbon payments. Carbon enterprise investment was primarily directed at improved forest management and agricultural extension, rather than at developing forest product value chains. However, forest product value chains — particularly honey and mushrooms — dominated both realised and aspirational income sources among CF communities. Most forest products are sold only at local markets. Partnerships with private enterprises were uncommon and mostly involved honey off-takers. Communities identified timber as the most important unrealised commercial opportunity, despite none of the surveyed CFMGs being involved in timber production. Opportunity costs of CFM were recognised by only about one third of FGD groups. Unexpectedly, non-timber forest product (NTFP) harvesting was frequently cited as a precluded activity, suggesting possible misunderstandings around forest use rules in many communities. It is evident from the survey that most CFMG business plans are overly reliant on carbon and potentially valuable NTFP value chains remain under-exploited. Furthermore, the governance index was a significant predictor of economic performance, at least from the perspective of the community FGD groups, which again underscores the importance of ensuring quality governance if the benefits of CFM are to be realised.

Key conclusions

CFM is generating economic benefits for communities. However, these are currently very limited and overly focused on carbon, too concentrated in a few low paying roles, such as HFOs, and too unevenly distributed across communities, with women in particular

not seeing sufficient economic benefits. To achieve an economic transformation through CFM, requires a diversified, layered forest enterprise model — one that exploits the full range of products, develops value chains and actively connects communities to markets. Such a vision is both achievable and arguably essential for long-term CFM viability, but requires a much greater effort from the FD and CFM partners, such as carbon enterprises, to identify opportunities and provide capacity development in forest resource management, value addition and business skills.

Introduction

In this chapter, we consider the effect of CFM management on indicators of economic transformation.

Access to natural resources

We asked FGD groups whether CFM had improved access to natural resources for household use (Figure 1).

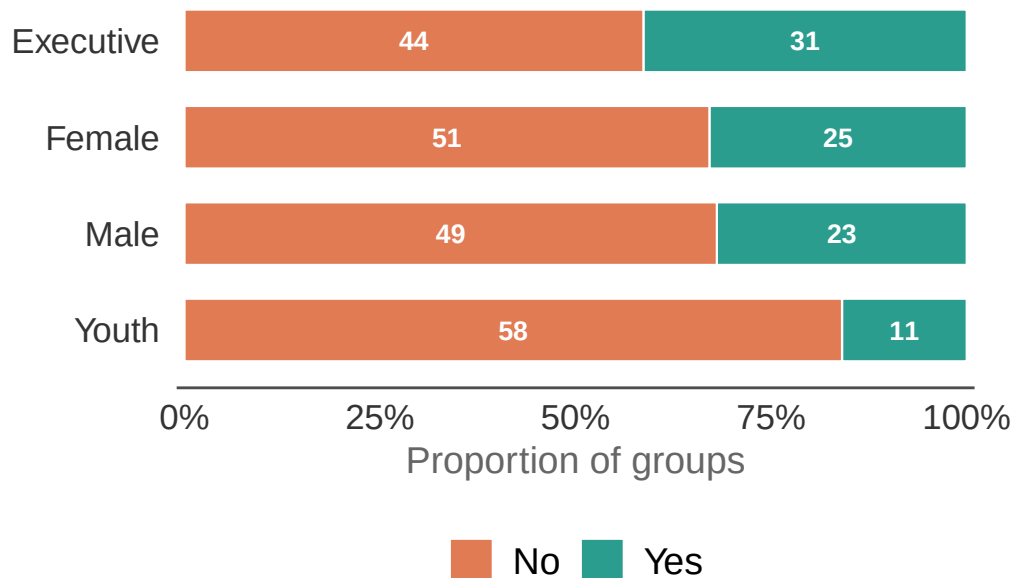


Figure 1: Has CFM improved access to natural resources for household use? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 1: Results of the model examining the effect of FGD type and governance index on whether communities report improved access to natural resources for household use.

Model results for improved access to natural resources for household use — FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.117; CF-within-province variance = 0.327. Marginal $R^2 = 0.087$; conditional $R^2 = 0.196$. $n = 273$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Executive FGD)	-1.060	0.461	0.0214
Female FGD	-0.433	0.370	0.2429
Male FGD	-0.471	0.379	0.2138
Youth FGD	-1.379	0.430	0.0013
Governance index	0.145	0.078	0.0633

Most FGD groups reported no increase in access to natural resources for household use, though approximately one third indicated that access had improved under CFM. Community youth were significantly more likely to report no improvement compared to CFMG executive FGDs, but none of the other pairwise comparisons were significant. Most natural resources are currently accessed for household use and results were similar for those that are collected for sale.

Individual interviewees were also asked whether CFM had improved access to natural resources for household use (Table 3). Similar to the FGD results, a majority of respondents indicated that they did not think that CFM had improved access to natural resources for household use. However, DFOs were significantly more likely to suggest that CFM had improved access than CFMG executives or traditional leaders. None of the other pairwise comparisons were significant.

Employment

We asked FGD groups whether CFM had generated employment opportunities (Figure 2).

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As can be seen from the figure, approximately half of the FGD groups agreed that CFM had increased employment opportunities. Interestingly, community youth were especially likely to agree that CFM had created employment opportunities. In contrast, community female FGD groups were much more likely to say that CFM had not increased employment opportunities. All pairwise comparisons were highly significant.

Table 2: Pairwise comparisons among FGD types for improved access to natural resources for household use.

Pairwise comparisons — FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	<i>P</i> value
Executive - Female	0.433	0.370	1.168	0.6472
Executive - Male	0.471	0.379	1.243	0.5993
Executive - Youth	1.379	0.430	3.209	0.0073
Female - Male	0.039	0.394	0.098	0.9997
Female - Youth	0.946	0.439	2.156	0.1359
Male - Youth	0.908	0.446	2.033	0.1757

Table 3: Percentage of individual interviewees reporting improved access to natural resources as an economic benefit of CFM.

Improved access to natural resources for household use as reported by individual interviewees

Values represent the % of responses agreeing that CFM has improved access to resources for household use. n = the total number of interview responses.

Role	n	Household use
Community member	31	45%
CFMG executive member	63	24%
Traditional leader	75	27%
District Forestry Officer	56	62%
Private enterprise rep	32	50%

Table 4: Results of the model examining the effect of interviewee role, gender and age on whether individuals report improved access to natural resources for household use.

Model results for improved access to natural resources for household use — individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.129; CF-within-province variance = 0.149. Marginal $R^2 = 0.122$; conditional $R^2 = 0.19$. $n = 257$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Community member)	0.235	0.810	0.7718
CFMG executive member	-1.023	0.527	0.0520
Traditional leader	-0.823	0.490	0.0934
District Forestry Officer	0.695	0.502	0.1664
Private enterprise rep	0.098	0.555	0.8599
Male	-0.317	0.372	0.3943
Age	-0.002	0.013	0.8479

Table 5: Pairwise comparisons among interviewee roles for improved access to natural resources for household use.

Pairwise comparisons — interviewee role

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	<i>P</i> value
cfmg_member - cfmg_exec_member	1.023	0.527	1.943	0.2946
cfmg_member - trad_leader	0.823	0.490	1.678	0.4477
cfmg_member - dfo	-0.695	0.502	-1.384	0.6380
cfmg_member - NGO_rep	-0.098	0.555	-0.177	0.9998
cfmg_exec_member - trad_leader	-0.200	0.460	-0.435	0.9926
cfmg_exec_member - dfo	-1.718	0.474	-3.628	0.0027
cfmg_exec_member - NGO_rep	-1.121	0.515	-2.175	0.1892
trad_leader - dfo	-1.518	0.453	-3.347	0.0073
trad_leader - NGO_rep	-0.921	0.515	-1.788	0.3807
dfo - NGO_rep	0.597	0.480	1.245	0.7250

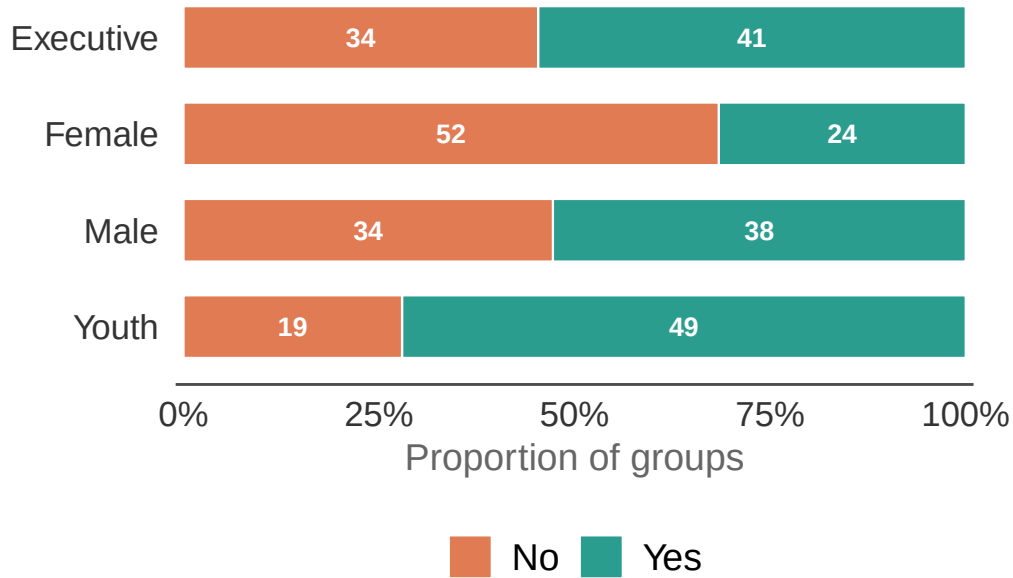


Figure 2: Responses to whether CFM has generated employment opportunities. The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 6: Results of the model examining the effect of FGD type and governance index on whether communities report that CFM has generated employment opportunities.

Model results for employment opportunities generated by CFM — FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.893; CF-within-province variance = 0. Marginal $R^2 = 0.287$; conditional $R^2 = 0.439$. $n = 272$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Executive FGD)	-1.617	0.566	0.0043
Female FGD	-1.459	0.419	0.0005
Male FGD	-0.384	0.400	0.3373
Youth FGD	1.124	0.427	0.0085
Governance index	0.434	0.082	0.0000

Table 7: Pairwise comparisons among FGD types for whether CFM has generated employment opportunities.

Pairwise comparisons — FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	<i>P</i> value
Executive - Female	1.459	0.419	3.484	0.0028
Executive - Male	0.384	0.400	0.959	0.7725
Executive - Youth	-1.124	0.427	-2.633	0.0421
Female - Male	-1.076	0.429	-2.508	0.0587
Female - Youth	-2.583	0.473	-5.457	0.0000
Male - Youth	-1.508	0.445	-3.387	0.0039

Individual interviewees were also asked whether CFM had generated employment opportunities (Table 8). Interestingly, community respondents were much more likely to agree that CFM had created employment opportunities than private enterprise representatives or DFOs. The differences between DFOs and CFMG executives and traditional leaders were significant, but none of the other pairwise comparisons were significant.

Full-time employment by gender, age group and province

CFMG Executive FGDs reported on the number of males, females, and youth in full-time employment in the CFMG (Figure 3). Clearly, there are many more men in full-time employment than women. According to the responses across the 75 CFMGs surveyed, 278 men were employed compared to 56 women, and 207 male youths compared to 48 female youths.

Types of formal employment opportunities

Community female, male, and youth FGDs identified the types of formal employment opportunities generated by CFM (Figure 4). Clearly, by far the largest type of job created is HFO or Community Scout.

Income earning opportunities

Carbon finance

We asked the CFMG executive FGDs whether their CFMG had received income from carbon finance (Figure 5). According to the responses, around one third of CFs have received carbon payments. This is an under-representation for the number of CFMGs involved in carbon

Table 8: Percentage of individual interviewees reporting that CFM has generated employment opportunities by interviewee role.

Employment opportunities reported by individual interviewees

Values represent the % of responses agreeing that CFM has generated employment opportunities. n = the total number of interview responses.

Role	n	Agree CFM has generated employment opportunities
Community member	31	60%
CFMG executive member	63	75%
Traditional leader	75	53%
District Forestry Officer	56	38%
Private enterprise rep	32	48%

Table 9: Results of the model examining the effect of interviewee role, gender and age on whether individuals report that CFM has generated employment opportunities.

Model results for employment opportunities generated by CFM — individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.315; CF-within-province variance = 0.195. Marginal $R^2 = 0.116$; conditional $R^2 = 0.235$. n = 250.

Factor	Estimate	SE	<i>P</i> value
Intercept (Community member)	2.211	0.922	0.0165
CFMG executive member	0.331	0.564	0.5572
Traditional leader	-0.088	0.531	0.8683
District Forestry Officer	-1.448	0.580	0.0125
Private enterprise rep	-0.975	0.634	0.1242
Male	0.436	0.380	0.2511
Age	-0.041	0.014	0.0031

Table 10: Pairwise comparisons among interviewee roles for whether CFM has generated employment opportunities.

Pairwise comparisons — interviewee role

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	<i>P</i> value
cfmg_member - cfmg_exec_member	-0.331	0.564	-0.587	0.9770
cfmg_member - trad_leader	0.088	0.531	0.166	0.9998
cfmg_member - dfo	1.448	0.580	2.498	0.0909
cfmg_member - NGO_rep	0.975	0.634	1.537	0.5381
cfmg_exec_member - trad_leader	0.419	0.463	0.904	0.8953
cfmg_exec_member - dfo	1.779	0.484	3.677	0.0022
cfmg_exec_member - NGO_rep	1.306	0.552	2.366	0.1248
trad_leader - dfo	1.360	0.471	2.890	0.0315
trad_leader - NGO_rep	0.887	0.543	1.632	0.4767
dfo - NGO_rep	-0.473	0.502	-0.942	0.8806

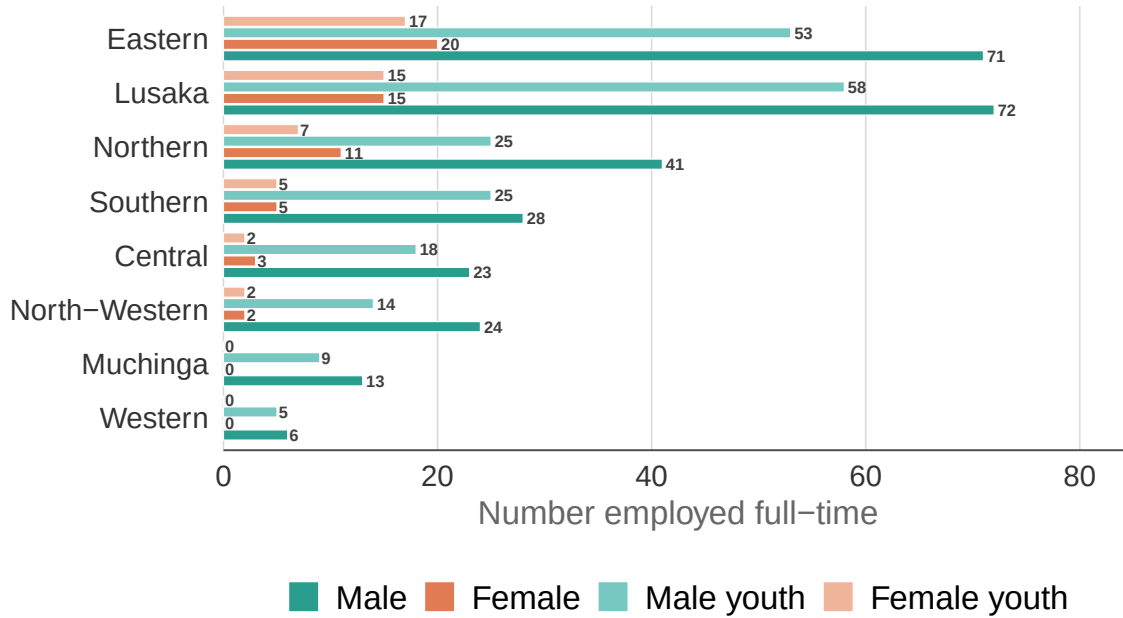


Figure 3: Number of males, females, male youth, and female youth employed full-time in CFMGs, by province, as reported by executive FGDs.



Figure 4: Types of formal employment opportunities reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting that job type.

finance, which is around 80% of CFMGs, and reflects only those that have actually started to receive payments.

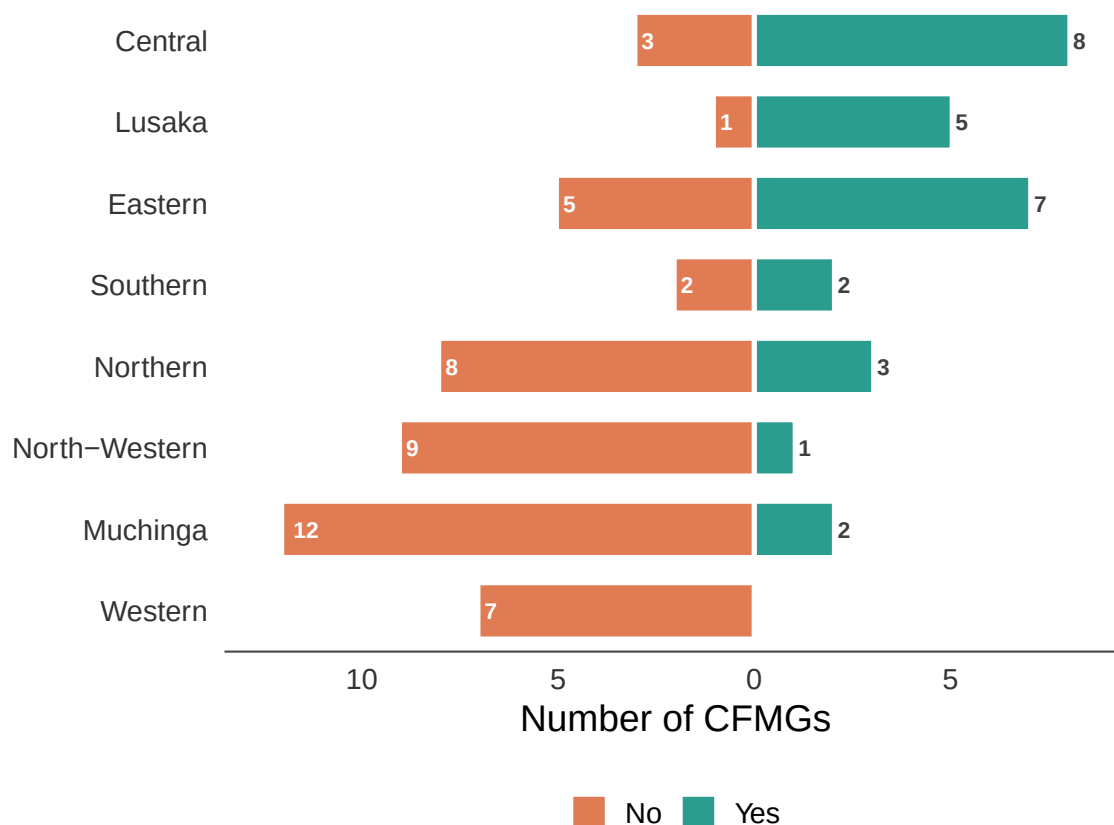


Figure 5: CFMG involvement in carbon finance by province, as reported by CFMG executives.

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FGD groups were also asked whether the community was aware of the carbon credit agreement (Figure 6). While most CFMG executive groups stated that the community was aware of the carbon finance agreement, interestingly two stated that they were not. Most community female and male groups indicated they were aware of the carbon finance agreement, but more than half of the community youth groups stated they were not aware of the carbon finance agreement. Community youth were significantly less likely to know about the carbon finance agreement than CFMG Executives, but other pairwise comparisons were not significant.

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Individual interviewees were also asked whether the community was aware of the carbon credit agreement (Table 13). Most respondents indicated that the community was aware, but the

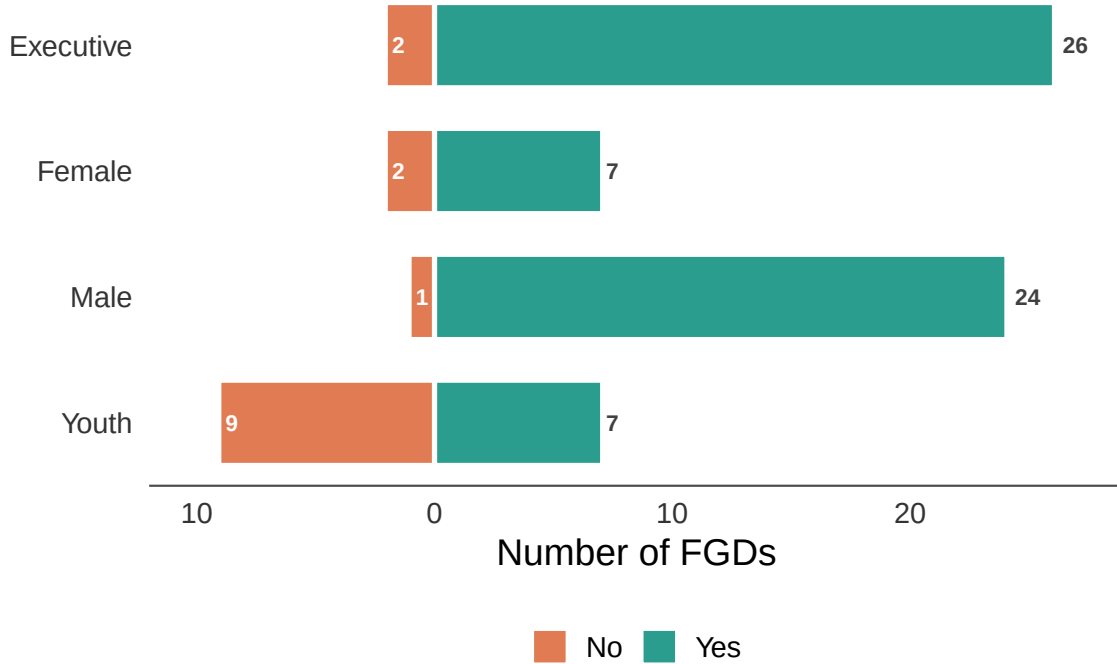


Figure 6: Community awareness of the carbon credit agreement by FGD group.

Table 11: Results of the model examining the effect of FGD type and governance index on community awareness of the carbon credit agreement.

Model results for community awareness of carbon credit agreement — FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0; CF-within-province variance = 0. Marginal $R^2 = 0.987$; conditional $R^2 = \text{NA}$. $n = 70$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Executive FGD)	3.499	1.214	0.0039
Female FGD	-1.681	1.127	0.1359
Male FGD	33.852	14,779,767.795	1.0000
Youth FGD	-2.659	0.907	0.0034
Governance index	-0.189	0.184	0.3051

Table 12: Pairwise comparisons among FGD types for community awareness of the carbon credit agreement.

Pairwise comparisons — FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	<i>P</i> value
Executive - Female	1.681	1.127	1.491	0.4428
Executive - Male	-33.852	14,779,767.795	0.000	1.0000
Executive - Youth	2.659	0.907	2.932	0.0177
Female - Male	-35.533	14,779,767.795	0.000	1.0000
Female - Youth	0.979	1.002	0.976	0.7629
Male - Youth	36.511	14,779,767.795	0.000	1.0000

percentage of positive responses varied across interviewee roles. Interestingly, private enterprise representatives overwhelmingly (~95%) suggested the community was informed, but the only 70% of DFOs thought the communities were informed, although the difference was not significant.

Carbon enterprise support

The types of CFM activities supported by carbon enterprises are shown in Figure 7. As can be seen, support focused on aspects of forest management or agricultural extension. Other potential elements of support, such as NTFPs value chains, infrastructure development or market access, featured substantially less.

Other income earning opportunities

FGD groups also reported whether CFM had generated income earning opportunities through specific interventions, including market linkages, alternative livelihood interventions, conservation revenue, forest product processing and ecotourism (Figure 8). Although some community FGDs indicated improvement in income earning opportunities under some categories, especially conservation revenue and forest product processing, most indicated that CFM had not generated income earning opportunities in these areas.

However, the community groups did identified other additional sources of income generated by CFM (Figure 9). Natural resource based value chains featured prominently among these, but interestingly community business was also mentioned.

To better understand the importance of different income earning opportunities we asked the community female, male, and youth FGD groups to rank their top three additional sources of

Table 13: Percentage of individual interviewees reporting community awareness of the carbon credit agreement, by interviewee role.

Community awareness of the carbon credit agreement, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Aware of carbon credit agreement
Community member	31	83%
CFMG executive member	63	87%
Traditional leader	75	80%
District Forestry Officer	56	70%
Private enterprise rep	32	95%

Table 14: Results of the model examining the effect of interviewee role, gender and age on community awareness of the carbon credit agreement.

Model results for community awareness of carbon credit agreement — individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0; CF-within-province variance = 0. Marginal $R^2 = 0.94$; conditional $R^2 = \text{NA}$. n = 97.

Factor	Estimate	SE	<i>P</i> value
Intercept (Community member)	19.333	6,639.257	0.9977
CFMG executive member	0.305	0.981	0.7554
Traditional leader	-0.038	1.070	0.9716
District Forestry Officer	-0.456	0.985	0.6435
Private enterprise rep	1.645	1.336	0.2183
Male	-19.029	6,639.257	0.9977
Age	0.020	0.029	0.4943



Figure 7: Types of CFM activities supported by the carbon trader.

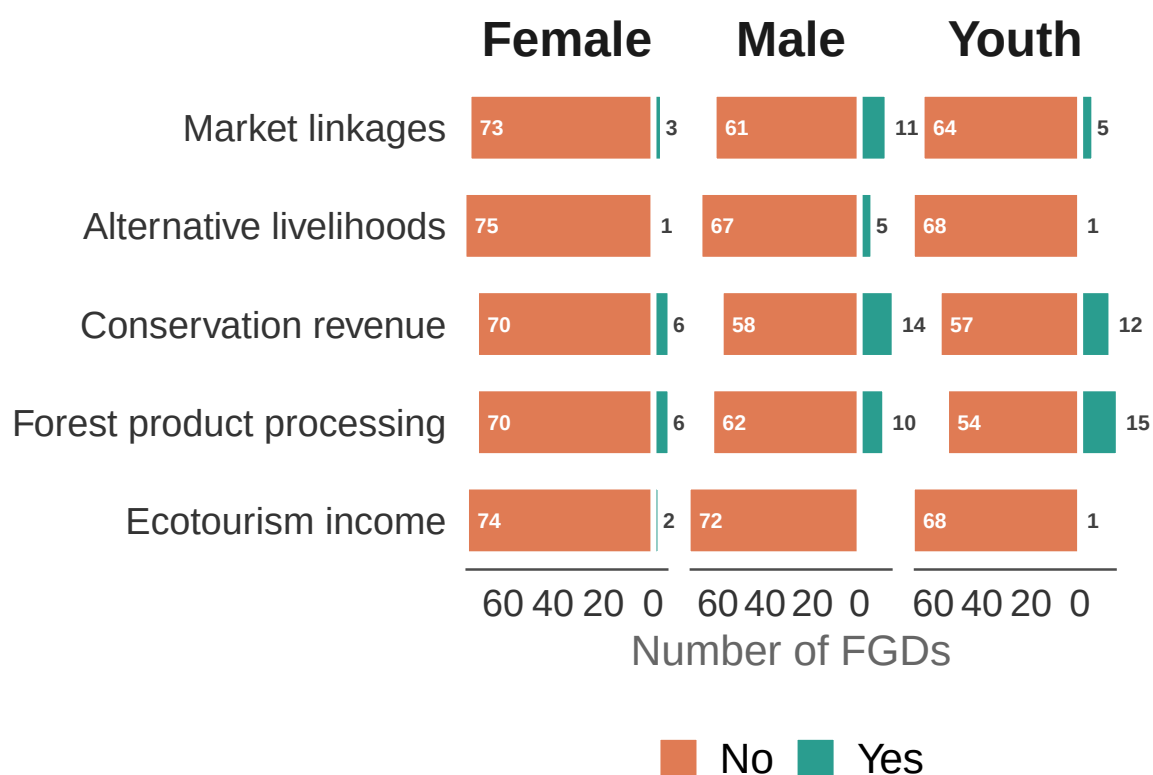


Figure 8: Income earning opportunities generated by CFM, as reported by female, male, and youth FGDs. The length of the bars represents the number of FGDs reporting a benefit (Yes) or no benefit (No) for each income source.



Figure 9: Additional sources of income from CFM reported by female, male, and youth FGDs.
Word size is proportional to the number of FGDs reporting each income source.

income from CFM (Figure 10). Interestingly, among the 14 value chains identified, 12 were mentioned as 1st rank at least once. This immediately points to the diversity of forest products being harvested from CFs across the country, but also to the fact that for most CFs only one product features strongly as an additional source of income. Among these, mushrooms and honey are clearly the most frequently mentioned.

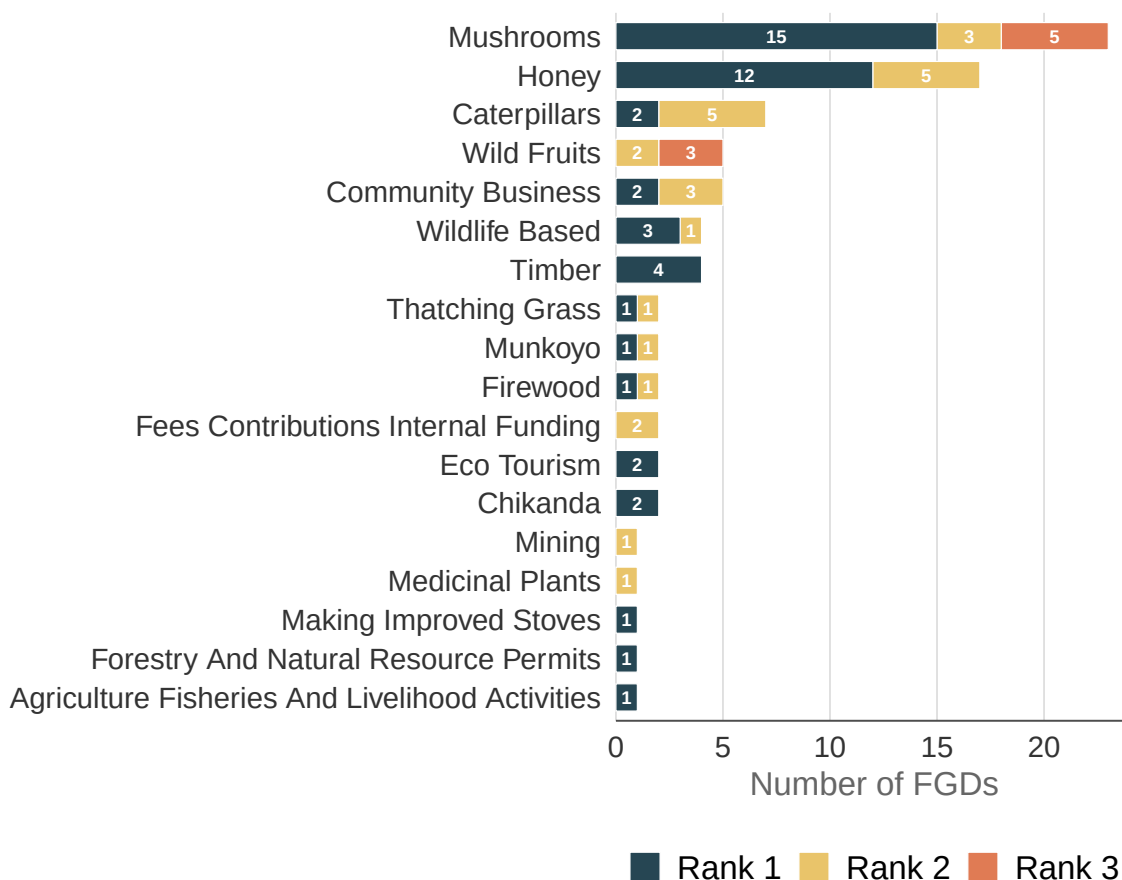


Figure 10: Ranking of additional income sources from CFM by female, male, and youth FGDs. Bars show the number of FGDs assigning each rank. Only income sources receiving at least one top-three ranking are shown.

Female, male, and youth FGDs also reported which forest products were harvested for household use (Figure 11). Mushrooms again featured prominently, but other natural resources are also clearly important, such as wild fruits, medicinal plants, thatching grass and caterpillars.

CFMG executive FGDs were asked whether the CFMG was collaborating with any private enterprises (Figure 12). The answers indicate that only a small number of CFMGs were collaborating with a private enterprise.

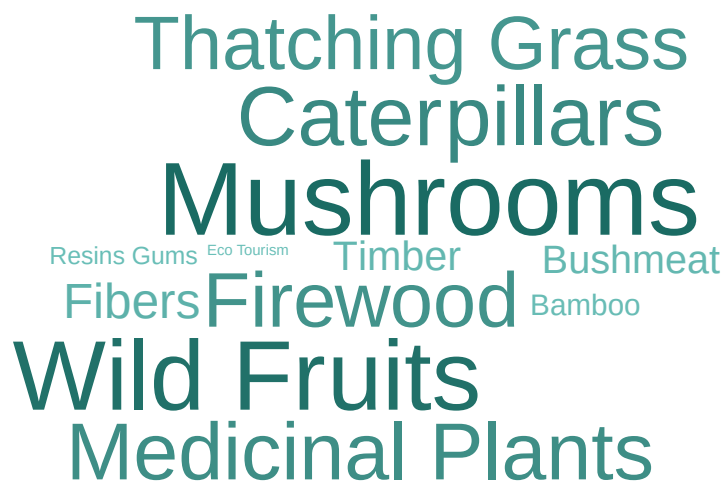


Figure 11: Forest products harvested for household use reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each product.

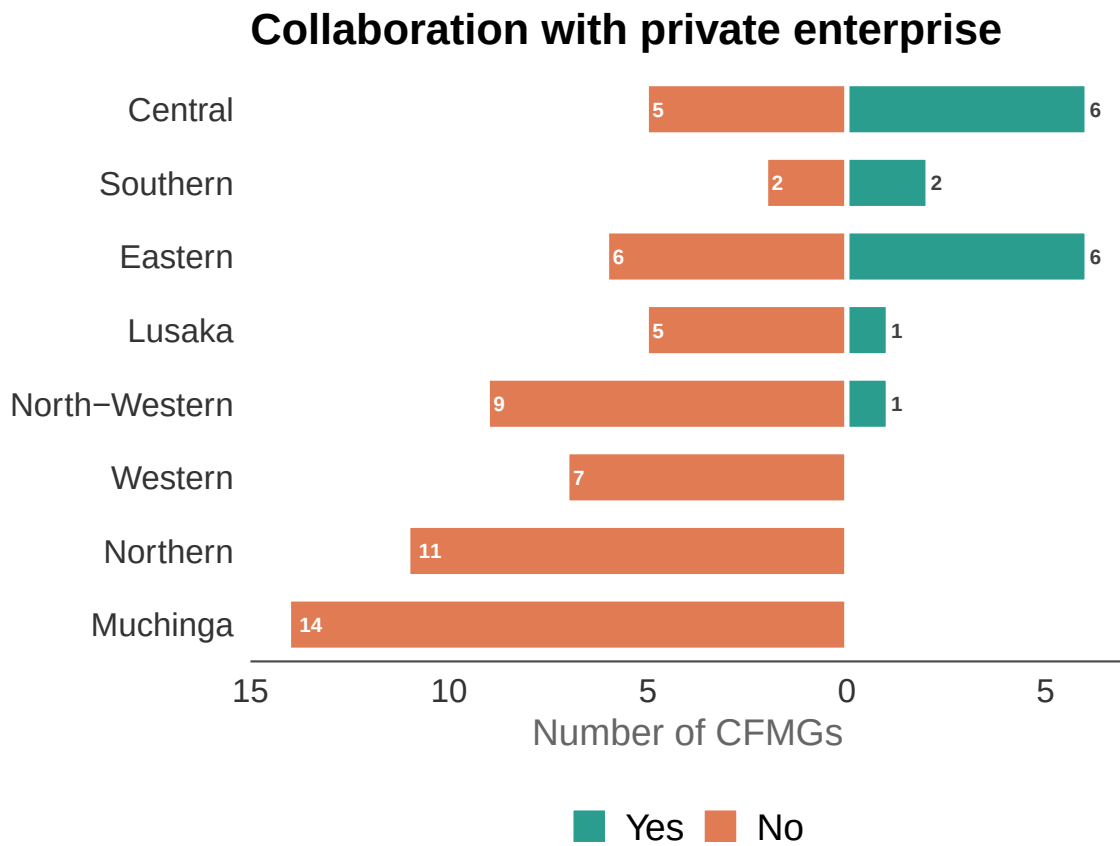


Figure 12: CFMG collaboration with private enterprise as reported by executive FGDs by province.

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The value chains that CFMGs reported working with through private enterprise partnerships are shown in Figure 13. Sixteen out of 17 involved partnerships with honey off-takers and just one involved mushrooms.

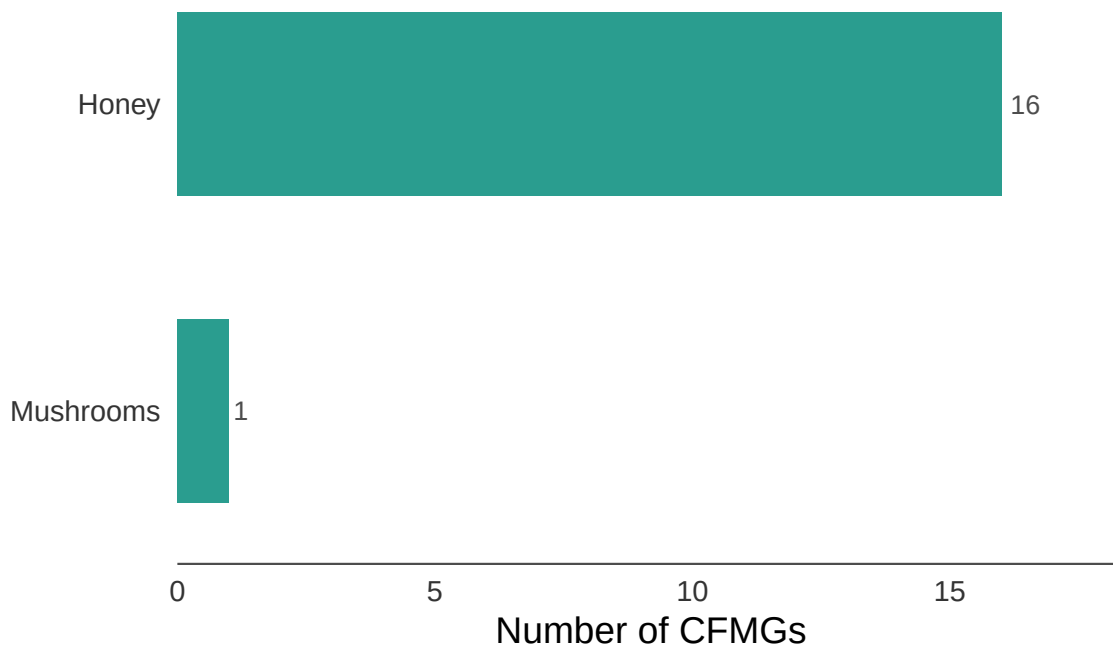


Figure 13: Value chains that CFMGs collaborate with through private enterprise partnerships, as reported by executive FGDs.

We asked the community female, male, and youth FGD groups to identify economic opportunities that remained unrealised within their CFs (Figure 14). Interestingly, although a wide range of different natural resource related value chains were mentioned, timber was mentioned the most often. This is noteworthy as none of the CFMGs we surveyed are involved in timber production.

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The ranking of these unrealised opportunities is shown in Figure 15. Again a wide range of resources are mentioned, but the top three are timber, honey and mushrooms. Again, it is interesting that community groups consider their timber resources to have high unrealised potential for commercial development.

FGD groups also identified potential value-added opportunities from CFM resources (Figure 16). The overlap with the previous two questions is substantial, but it is interesting that

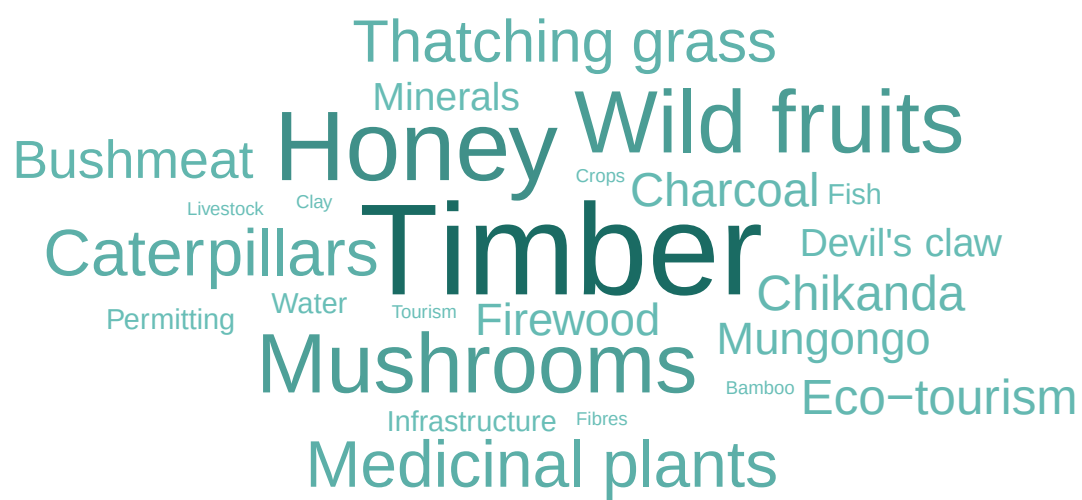


Figure 14: Unrealised economic opportunities identified by FGD groups.

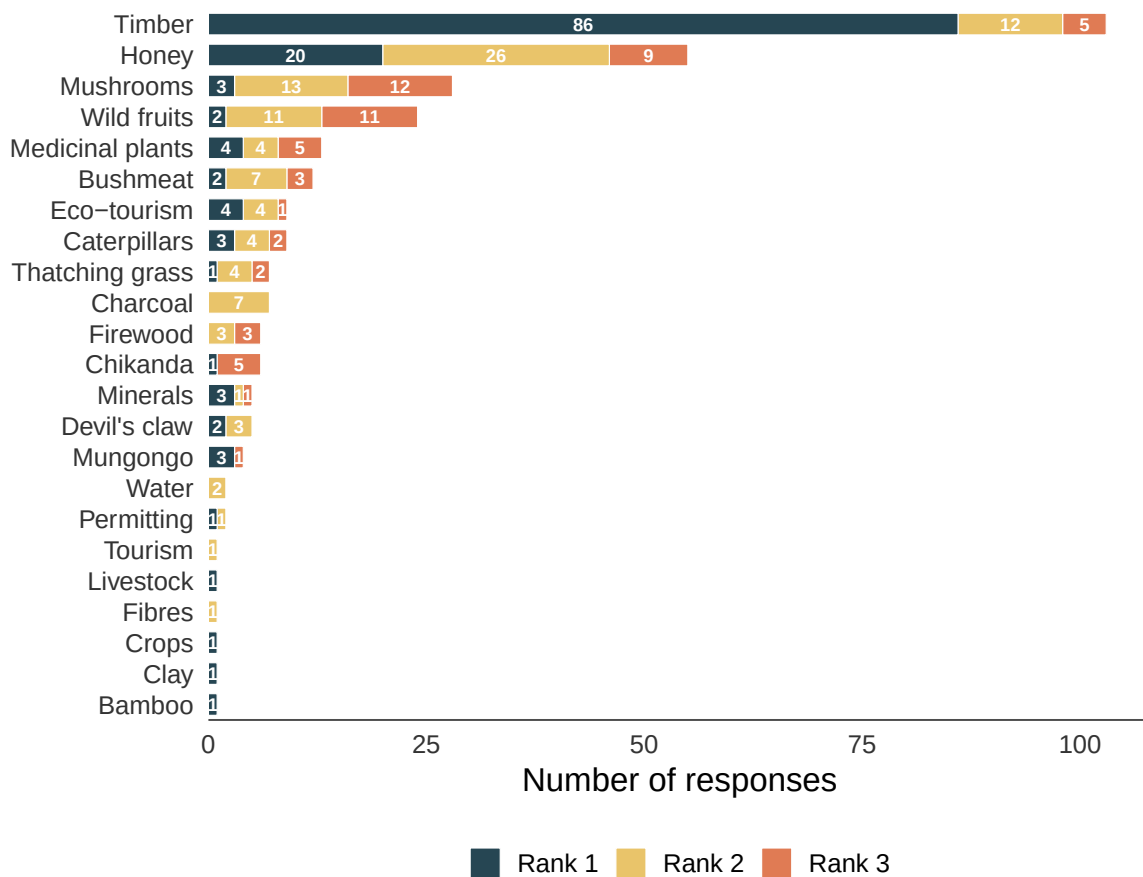


Figure 15: Ranking of unrealised economic opportunities by FGD groups. Bars show the frequency of rank 1 (highest priority), rank 2, and rank 3 responses.

the communities specifically identify honey processing and drying mushrooms, as well as timber processing. This suggests that the communities are aware that there are value addition opportunities that they are currently not realising.



Figure 16: Potential value-added opportunities identified by FGD groups.

We asked the CFMG Executive FGDs to report which primary markets CFM products are sold into (Figure 17). Evidently, most products are sold only at the local market level, with some also being transported to national markets.

Opportunity costs

Precluded activities

The community female, male, and youth FGD groups identified economic activities that are precluded under CFM regulations (Figure 18). While many of these are self evident, the fact that NTFP harvesting is listed quite prominently is surprising. Whether this reflects a misunderstanding on the part of the communities or indeed that they are being told they cannot collect NTFPs is not clear.

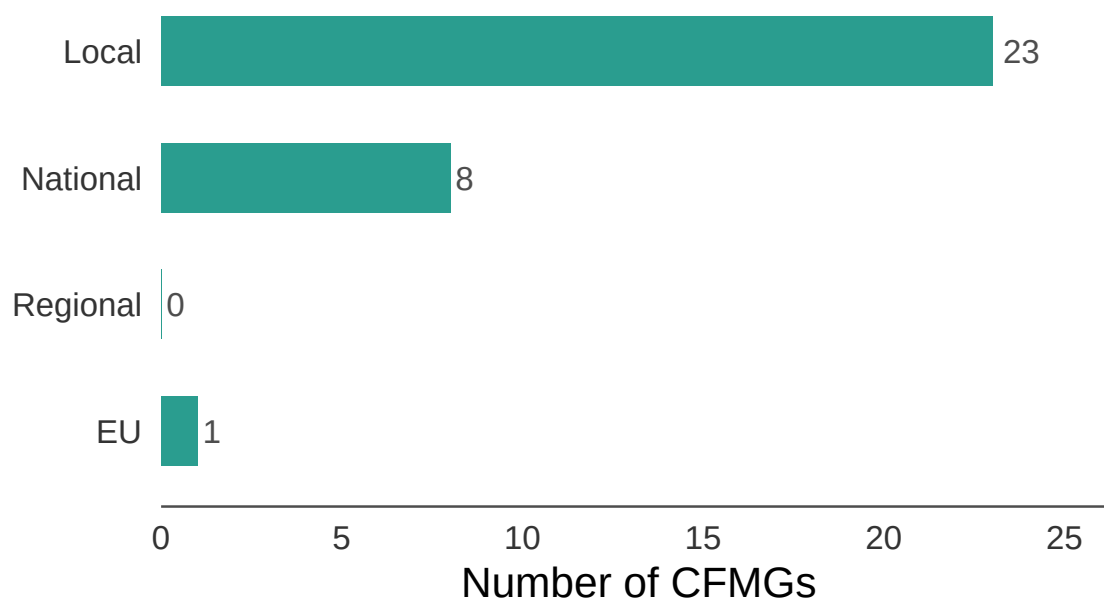


Figure 17: Primary markets for CFM products, as reported by executive FGDs.

Other opportunity costs

FGD groups were asked whether CFM entails opportunity costs for the community (Figure 19). Interestingly, many of the FGDs did not recognise any opportunity costs. However, about a third of the FGDs did. The proportion of FGDs recognising opportunity costs was highest among community male FGDs and lowest among community female FGDs. However, none of the pairwise comparisons were significant.

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Individual interviewees were also asked whether CFM entails opportunity costs. Table 16 shows the percentage reporting this broken down by interviewee role.

The types of opportunity costs identified are shown in Figure 20. Mentions of reduced access, land restrictions and land use conflict are by some margin the most frequently identified opportunity costs. However, it is also interesting that time costs and diverting community resources are also quite prevalent. CFM requires that community members meet which entails a time cost and community resources are diverted to managing the forest that might otherwise be spent on other activities, such as agriculture.

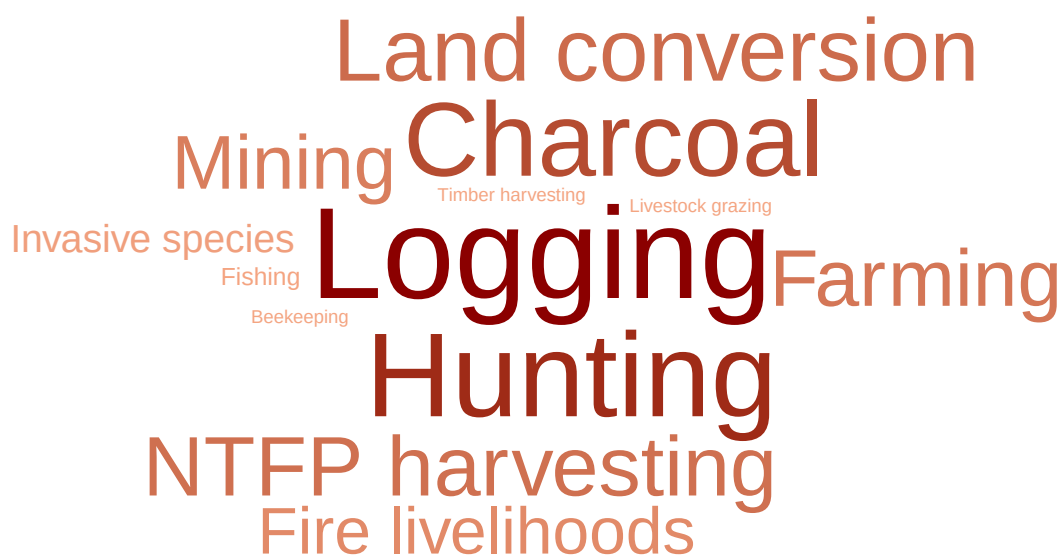


Figure 18: Economic activities precluded by CFM, as reported by FGD groups.

Table 15: Results of the model examining the effect of FGD type and governance index on whether communities report opportunity costs of CFM.

Model results for opportunity costs of CFM — FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.867; CF-within-province variance = 0.049. Marginal $R^2 = 0.041$; conditional $R^2 = 0.25$. $n = 271$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Executive FGD)	-0.890	0.542	0.1003
Female FGD	-0.515	0.383	0.1791
Male FGD	0.322	0.374	0.3884
Youth FGD	-0.441	0.385	0.2514
Governance index	0.122	0.072	0.0934

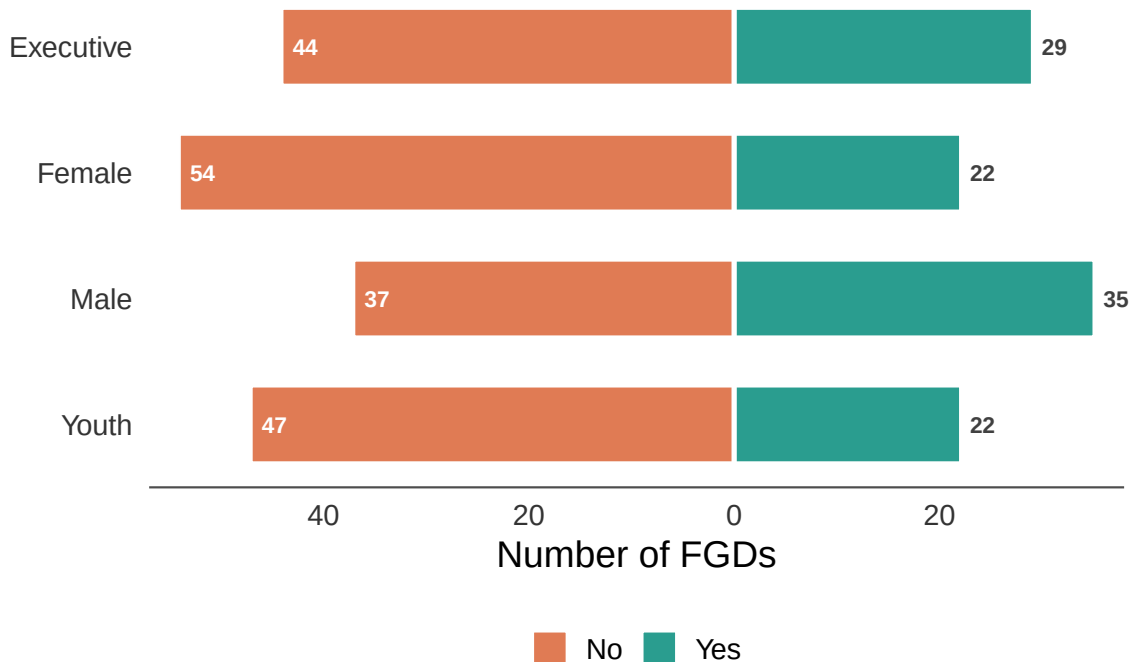


Figure 19: Whether CFM entails opportunity costs for the community, by FGD group.

Table 16: Percentage of individual interviewees reporting that CFM entails opportunity costs for the community, by interviewee role.

Opportunity costs of CFM, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses. A positive response indicates that the respondent considers CFM to entail opportunity costs for the community.

Role	n	Report opportunity costs
Community member	31	47%
CFMG executive member	63	51%
Traditional leader	75	35%
District Forestry Officer	56	39%
Private enterprise rep	32	59%

Table 17: Results of the model examining the effect of interviewee role, gender and age on whether individuals report opportunity costs of CFM.

Model results for opportunity costs of CFM — individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.385; CF-within-province variance = 0. Marginal R^2 = 0.033; conditional R^2 = NA. n = 245.

Factor	Estimate	SE	<i>P</i> value
Intercept (Community member)	-0.955	0.796	0.2301
CFMG executive member	-0.109	0.471	0.8168
Traditional leader	-0.344	0.470	0.4649
District Forestry Officer	-0.203	0.502	0.6858
Private enterprise rep	0.721	0.566	0.2032
Male	0.405	0.356	0.2557
Age	0.007	0.012	0.5358



Figure 20: Types of opportunity costs experienced under CFM.

Economic transformation index

To summarise the overall economic transformation performance of each CFMG, we constructed an economic transformation score (0–5) by summing five binary indicators reported by FGD groups — improved access to natural resources for household use, new employment opportunities created, carbon payments received (executive FGDs only), collaboration with a private enterprise (executive FGDs only), and absence of opportunity costs — scored as 1 (positive outcome reported) or 0 (no positive outcome or worsening). The distribution of economic transformation scores across CFMGs is shown in Figure 21 (Appendix III). We then modelled this score using a linear mixed model with the governance index as the predictor, and random intercepts for province and CF within province, to test whether governance was a significant predictor of overall economic transformation performance (Table 18, Figure 22).

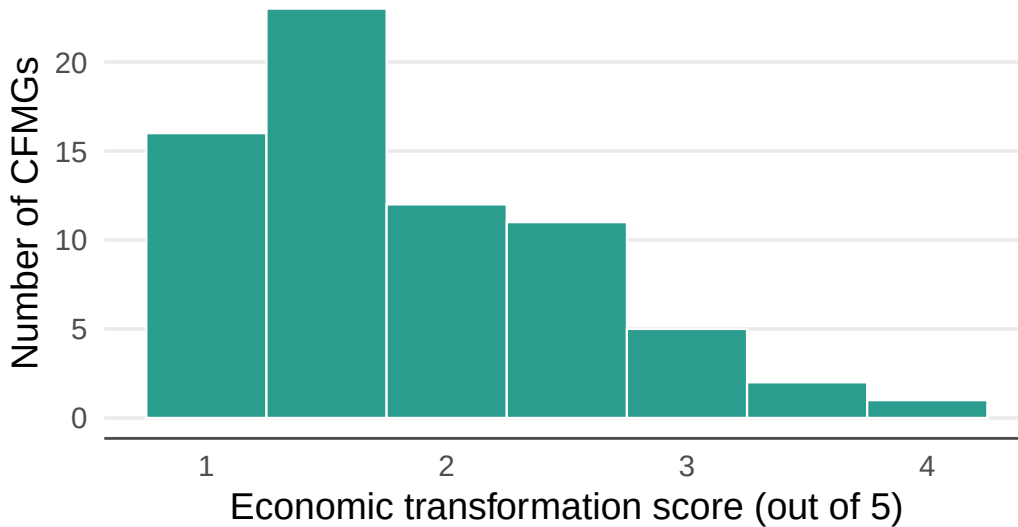


Figure 21: Distribution of mean economic transformation scores across CFMGs (0–5). Each bar represents the number of CFMGs with that mean score, averaged across all FGD types. The score sums five binary indicators: improved household resource access, new employment opportunities, carbon payments received, collaboration with a private enterprise, and absence of opportunity costs.

Discussion

Through the FGDs and the individual interviews, we investigated a diverse set of indicators of economic transformation. We covered access to natural resources for household use and sale, employment, income earning opportunities, including carbon finance and other sources of income, and opportunity costs.

Table 18: Results of the model examining the effect of the governance index on economic transformation performance.

Model results for economic transformation performance — governance index

Linear mixed model with random intercepts for province and CF within province. Province variance = 0.085; CF-within-province variance = 0.128; residual variance = 0.853. Marginal $R^2 = 0.071$; conditional $R^2 = 0.257$. $n = 273$.

Factor	Estimate	SE	<i>P</i> value
Intercept	1.363	0.204	0.0000
Governance index	0.135	0.036	0.0003

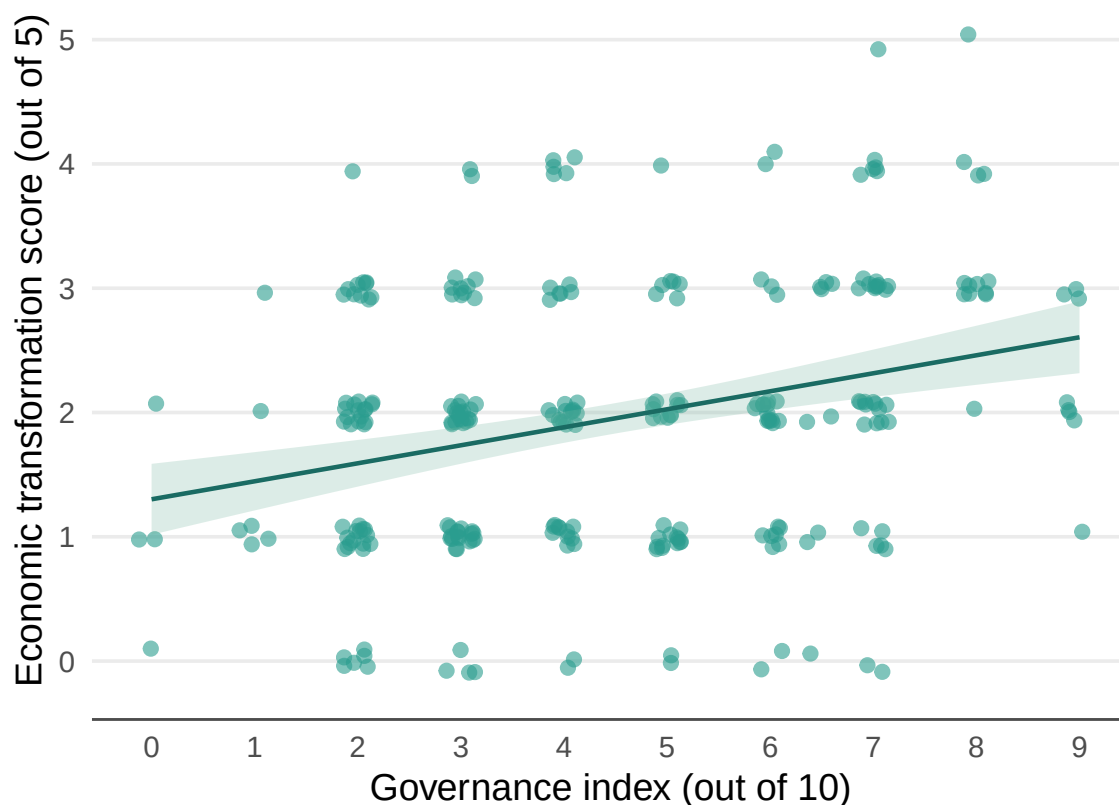


Figure 22: Relationship between the CFMG governance index and economic transformation score. Each point represents one FGD session. The line shows the fitted relationship from the linear mixed model, with the shaded band representing ± 1 SE.

Most FGD groups and individual interviewees indicated that CFM had not increased access to natural resources for either household use or sale. However, given that communities have customary rights of access and were involved in harvesting forest products before the instigation of CFM, it is not surprising that most respondents do not recognise any change. Nevertheless, around one third of respondent groups and individuals felt that access had improved under CFM, presumably reflecting formalisation of community rights to access forest resources.

A larger proportion of FGD groups considered that CFM had had a positive effect on full-time employment. However, the differences among FGD types were highly significant. More community female groups suggested there was no change in employment and more community youth groups felt full-time formal employment opportunities had increased. We also showed that the numbers employed included many more men than women, which would explain these differences in perception. Moreover, while CFM does appear to have increased formal employment, the quality of jobs was poor — most being HFOs or community scouts — and the numbers are paltry. In part, this may reflect the fact that employment in forest product harvesting and processing is rarely formal or full-time but, as we also found, there has been very limited development of forest resource based enterprises. Thus, while CFM does appear to have resulted in some new employment opportunities, these have been largely limited to low paying jobs and have not benefited many women.

A large proportion of CFs in Zambia were formed on the promise of carbon dollars. However, only approximately one third of the CFs in our sample appeared to have benefited from carbon payments at the time of the survey. This number reflects the actual disbursement of carbon derived income to the community, as there clearly has been investment in communities by carbon enterprises. The fact that a number of the FGD groups, especially youth groups, were unaware of the carbon agreement is concerning. It suggests that the carbon enterprises may have failed to conduct a proper FPIC process involving the entire community and that the CFMG executives may not be communicating properly with their communities. Nevertheless, many carbon enterprises are clearly supporting communities. Most activities concern forest protection and agricultural extension for sustainable intensification, such as climate smart agriculture. Both of which may be considered as essential to a carbon business plan. However, NTFPs and forest resource based enterprise development were far less often mentioned. We suggest this is an oversight on the part of carbon enterprises. Enhancing forest product value chains would increase the value of forest conservation for local communities. Furthermore, carbon finance alone is unlikely to fund development in forest margin communities sufficiently. Hence, there is a significant risk that communities will become disillusioned with carbon finance, especially among sections of the community that feel there is an inequitable sharing of benefits. Investing in forest based enterprises should be considered essential to a carbon business plan.

The above results are backed up by the response to questions concerning other income earning opportunities. It seems CFM has not led to increased income from market linkages, alternative livelihoods, conservation revenue, forest product processing or ecotourism except for in a very few cases. Nevertheless, a wide variety of forest products clearly contribute to household use

and some additional income. Although honey and mushrooms were most often identified as the most important source of additional income, another ten NTFPs were mentioned as the most important at least once. However, these additional sources of income were identified in less than half of the CFs and only 13 CFs identified a second ranked forest product. Thus, forest products are rarely being exploited as a source of additional income and if they are it is usually just one. Only a small number of CFMGs (23%) are partnering with an enterprise, such as an off-taker, and among those that do it almost exclusively concerns honey. The communities identified a wide diversity of forest products that could potentially be harvested and processed to provide an additional source of income. Prominent among these were timber and wild fruit, as well as mushrooms and honey. In some cases timber harvesting may conflict with carbon business plans, but not if plantations were established. The potential of CFM to support forest product enterprises is clearly massively under realised.

To get a more accurate picture of forest product values a more in depth PEN type survey, involving surveys of household forest product use repeated multiple times over a year, is recommended. The intensity of the survey would necessitate sub-sampling communities from our survey. However, the results of our survey could be used to stratify sampling, so that results could be extrapolated to the national level.

The local communities identified a number of opportunity costs to CFM. Most of these pertained to activities that are precluded, such as those involving forest clearance. However, many groups also identified other opportunity costs, such as time costs and diverted resources. Recognition of opportunity costs is important to understand the cost-benefits of CFM for communities, especially as these costs are likely to be unevenly distributed among community members. It was concerning that many groups mentioned restricted access to NTFPs. Whether this results from a misunderstanding of the regulations surrounding CFM or is because community members are actually being barred from harvesting NTFPs needs to be resolved. CFM should facilitate access to resources and improve their management, not prevent access.

It is noteworthy that the governance index was a highly significant predictor of our economic transformation index. Thus, as with our results from the chapter on SFM, performance is predicated on good governance. It is imperative that the FD and other CFMG partners step up their oversight of CFMG governance to ensure long term positive outcomes from CFM.

Overall, a positive picture of economic transformation emerges from the responses of the FGDs and individual interviews. While changes in indicators, such as employment or additional income from forest products, are not substantial, they are in a positive direction. In the future, most communities will also receive carbon income, even if this has only benefited around one third of CFs so far. However, it is also evident that CFMG business plans are over-reliant on carbon or sometimes carbon and honey. Substantial opportunities for developing much more diversified business plans exist. These could include a range of different NTFPs and potentially also timber or sustainable charcoal. In addition to simply increasing income earning opportunities, diversified business plans would also provide a buffer against fluctuating commodity prices and because different products are harvested at different times of the year, they can help stabilise income throughout the year.